

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2023	Date:	

Code of Conduct

I b.jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: jahnavi

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

1. Reference Resolution

Design an algorithm for **Reference Resolution** that identifies noun phrases and pronouns and links them to their appropriate antecedents using **recency, grammatical agreement, semantic compatibility, and discourse context**. Apply your algorithm to the following discourse:

"Meena gave Priya the project report after she completed the analysis. She asked her to review it before the meeting. The report contained several useful graphs."

Generate the **resolved discourse** by replacing ambiguous pronouns with their most likely antecedents. Further, explain how **gender agreement, syntactic role, recency, and discourse context** help resolve the ambiguities. Discuss one possible alternative interpretation and explain why your algorithm rejects it.

2. Text Coherence and Discourse Structure

Develop an algorithm to identify **discourse relations and coherence** among multiple sentences using linguistic connectives, semantic dependencies, temporal information, and causal relationships. Apply your algorithm to the following discourse:

"The research team collected a large dataset. The dataset contained several missing values. Therefore, the researchers performed data preprocessing. After cleaning the data, they trained the machine-learning model. The model achieved high prediction accuracy."

Identify the relationships such as **elaboration, problem-solution, cause-effect, temporal sequence, and result** among the sentences. Construct a **hierarchical discourse structure representation** and justify how each relation contributes to the overall coherence of the passage.

3. Dialogue Act Recognition

Design an algorithm for a **Dialogue Act Recognition System** that classifies conversational utterances based on **syntactic patterns, keywords, contextual information, and dialogue history**. Apply your algorithm to the following conversation:

User: "I need to change my flight reservation."

Agent: "Certainly. What date would you like to travel?"

User: "Please change it to Friday."

Agent: "Your reservation has been changed to Friday."

User: "Thank you for your help."

Classify each utterance as **Request, Question, Inform, Confirmation, Action, or Acknowledgement**. Generate the complete **dialogue-act sequence** and explain how the sequence enables the conversational agent to track the user's goal, identify missing information, and determine when the task has been successfully completed.

4. Natural Language Generation and Surface Realization

Design an algorithm for **Natural Language Generation (NLG)** that transforms a structured semantic representation into a grammatically correct natural-language sentence. Consider the following semantic input:

Action: Submit

Agent: Student

Object: Assignment

Recipient: Professor

Tense: Future **Time:**

Tomorrow

Perform the following steps:

1. Identify the required semantic roles.
2. Perform appropriate **lexical selection**.
3. Determine the syntactic structure of the sentence.
4. Apply **tense and agreement rules**.
5. Perform surface realization.

Generate the final natural-language sentence and explain how the system transforms the semantic representation into the final output. Finally, validate the generated sentence for **grammaticality, semantic consistency, and naturalness**.

5. Hybrid Machine Translation

Design an algorithm that combines **Interlingua-based Machine Translation, Statistical Machine Translation, and contextual disambiguation** to translate a source-language sentence into a target language. Apply your algorithm to:

"The bank approved the student's loan application."

Perform the following steps:

1. Analyze the source sentence and identify its **syntactic and semantic structure**.
2. Construct an **Interlingua representation** containing the entities, actions, and semantic roles.
3. Identify potentially ambiguous words and generate multiple candidate translations.
4. Apply **statistical/contextual scoring** to rank the candidate translations.
5. Select the most appropriate target-language sentence.
6. Produce the final translation.

Finally, evaluate how **semantic representation, contextual information, and statistical scoring** work together to resolve lexical ambiguity and improve the accuracy and fluency of the final translation.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudo code Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Reference Resolution — Discourse Entity Linking Algorithm

Problem Understanding

Input: a discourse containing noun phrases (NPs) and pronouns. Output: each pronoun linked to its most likely antecedent NP. Constraints/features to use: recency (how close a candidate antecedent is), grammatical agreement (gender/number match), semantic compatibility (role plausibility, e.g. who can "complete an analysis"), and discourse context (topic/focus continuity).

Algorithm (Pseudocode)

ALGORITHM ResolveReferences(Discourse D)

1. Sentences $\leftarrow$ Split(D)
2. Entities $\leftarrow$ [] // list of all NPs seen so far, with attributes
3. FOR each sentence S in Sentences:
4. NPs $\leftarrow$ ExtractNounPhrases(S)
5. FOR each NP in NPs:
6. Attributes(NP) $\leftarrow$ {gender, number, semanticRole, sentenceIndex}
7. Entities.append(NP)
8. Pronouns $\leftarrow$ ExtractPronouns(S)
9. FOR each pronoun P in Pronouns:
10. Candidates $\leftarrow$ Filter(Entities,
 matchGender(P) AND matchNumber(P))
11. FOR each candidate C in Candidates:
12. score(C) $\leftarrow$ $w_1 \cdot \text{RecencyScore}(C, P)$
 $+ w_2 \cdot \text{GrammaticalAgreementScore}(C, P)$
 $+ w_3 \cdot \text{SemanticCompatibilityScore}(C, P)$
 $+ w_4 \cdot \text{DiscourseSaliencyScore}(C, P)$
13. Best $\leftarrow$ ArgMax(Candidates, score)
14. Link(P, Best)
15. UpdateDiscourseFocus(Best)
16. RETURN ResolvedDiscourse

Application to the Given Discourse

Original: "Meena gave Priya the project report after she completed the analysis. She asked her to review it before the meeting. The report contained several useful graphs."

Resolved discourse: "Meena gave Priya the project report after Priya completed the analysis. Meena asked Priya to review it before the meeting. The report contained several useful graphs."

How the Features Resolve the Ambiguities

- Gender agreement: both "she" and "her" are feminine, so the candidate set is restricted to Meena and Priya only, ruling out any non-feminine entity.

- Syntactic role: "Priya" is the recipient/theme (indirect object) of "gave," and completing the analysis is more plausible for the person who was assigned the report, so the algorithm's semantic compatibility score favours Priya for the first "she."
- Recency: at the second sentence, "She asked her" — the subject "She" refers back to the most salient/topical entity from the previous clause structure (Meena, the giver/asker), while "her" (object) refers to Priya (the one being asked to act), consistent with the recipient role established earlier.
- Discourse context: once Meena is established as the agent who initiates actions (gave, asked) and Priya as the one who receives/performs them (completed, review), this role continuity keeps subsequent pronouns anchored to the correct entity across sentences.

Alternative Interpretation and Why It Is Rejected

An alternative reading could assign "she completed the analysis" to Meena instead of Priya (since Meena is the grammatical subject of the sentence). However, the algorithm rejects this because the semantic-compatibility score favours the entity who was just given the report (Priya) as the one who would go on to analyze it — this is the standard discourse pattern of "give X to Y, then Y acts on X" — giving Priya a higher combined score than Meena despite Meena's syntactic prominence as the main clause's subject.

Q2. Text Coherence and Discourse Structure Analysis Algorithm

Problem Understanding

Input: a multi-sentence passage. Output: the discourse relations that connect its sentences (elaboration, cause-effect, temporal sequence, problem-solution, result) and a hierarchical discourse-tree representation showing how the sentences combine to form a coherent passage.

Algorithm (Pseudocode)

ALGORITHM AnalyzeCoherence(Passage P)

1. Sentences <- Split(P)
2. Relations <- []
3. FOR i <- 1 to Length(Sentences)-1:
4. S1 <- Sentences[i]; S2 <- Sentences[i+1]
5. Connectives <- ExtractConnectives(S2) // e.g. "Therefore", "After", "Then"
6. SemanticOverlap <- ComputeEntityOverlap(S1, S2)
7. TemporalCue <- DetectTemporalMarker(S2)
8. CausalCue <- DetectCausalMarker(Connectives)
9. IF CausalCue present:
10. Relation <- "Cause-Effect"
11. ELSE IF TemporalCue present:
12. Relation <- "Temporal Sequence"
13. ELSE IF SemanticOverlap high AND NoNewProblemIntroduced(S2):
14. Relation <- "Elaboration"
15. ELSE IF ProblemDetected(S1) AND SolutionDetected(S2):
16. Relation <- "Problem-Solution"
17. ELSE IF OutcomeDetected(S2):
18. Relation <- "Result"
19. Relations.append((S1, S2, Relation))

```

20.     Tree <- BuildDiscourseTree(Sentences, Relations)
21. RETURN Tree, Relations

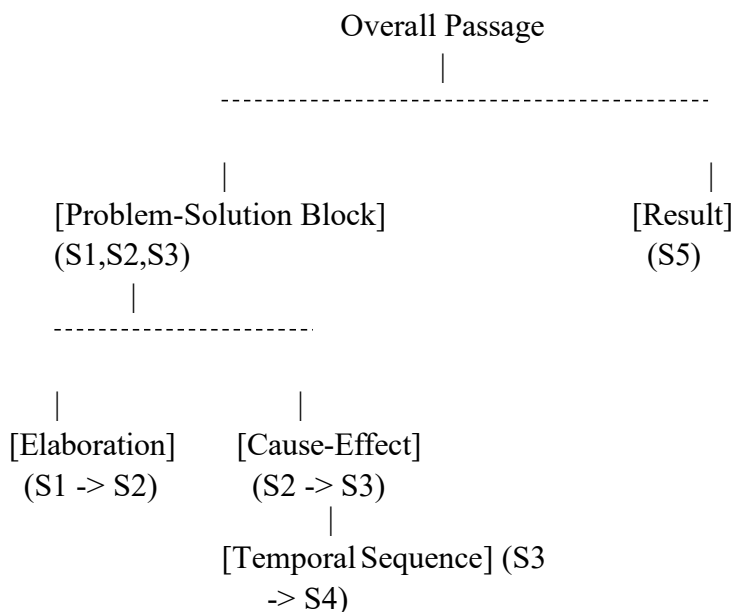
```

Application to the Given Passage

S1: The research team collected a large dataset. S2: The dataset contained several missing values. S3: Therefore, the researchers performed data preprocessing. S4: After cleaning the data, they trained the machine-learning model. S5: The model achieved high prediction accuracy.

- S1 → S2: Elaboration (S2 adds detail about the dataset introduced in S1).
- S2 → S3: Problem–Solution / Cause–Effect (missing values are the problem; "Therefore" signals preprocessing as the solution/effect).
- S3 → S4: Temporal Sequence ("After cleaning the data" marks a time-ordered next step).
- S4 → S5: Result (training leads to the outcome of high prediction accuracy).

Hierarchical Discourse Structure



Justification of Coherence

Each relation link ties a sentence to the sentence(s) needed to interpret it: elaboration keeps the dataset description grounded to its introduction; the cause-effect/problem-solution relation explains why preprocessing happened; the temporal marker orders preprocessing before training; and the result relation closes the passage by connecting the trained model back to the overall goal. Together these nested relations form a single connected discourse tree rather than a set of disjoint sentences, which is exactly what makes the passage coherent.

Q3. Dialogue Act Recognition — Intent Recognition Algorithm

Problem Understanding

Input: a sequence of conversational utterances from a user and an agent. Output: a dialogue-act label for each utterance from {Request, Question, Inform, Confirmation, Action, Acknowledgement}, using syntactic patterns, keywords, contextual/dialogue history.

Algorithm (Pseudocode)

ALGORITHM RecognizeDialogueAct(Utterance U, History H)

1. Tokens <- Tokenize(U)
2. IF ContainsQuestionMark(U) OR StartsWithWH_or_Aux(Tokens):
3. Act <- "Question"
4. ELSE IF ContainsImperativeVerb(Tokens) OR ContainsPhrase(U, {"I need", "please change", "co
5. Act <- "Request"
6. ELSE IF ContainsPhrase(U, {"thank you", "thanks", "great"}) AND NoNewInfo(U):
7. Act <- "Acknowledgement"
8. ELSE IF PriorAct(H) == "Request" AND ContainsPastTenseCompletionVerb(Tokens):
9. Act <- "Action" // agent reports a completed system action
10. ELSE IF PriorAct(H) == "Question" AND AnswersQuestion(U, H):
11. Act <- "Confirmation"
12. ELSE:
13. Act <- "Inform" // states a fact/detail
14. UpdateHistory(H, U, Act)
15. RETURN Act

Application to the Given Conversation

- U1 (User) : "I need to change my flight reservation." -> Request
U2 (Agent) : "Certainly. What date would you like to travel?" -> Question
U3 (User) : "Please change it to Friday." -> Request
U4 (Agent) : "Your reservation has been changed to Friday." -> Action
U5 (User) : "Thank you for your help." -> Acknowledgement

How the Sequence Enables Goal Tracking

The sequence Request → Question → Request → Action → Acknowledgement lets the agent maintain a goal state: the first Request sets the user's goal ("change reservation"); the agent's Question flags a missing slot (travel date) needed to fulfil that goal; the second Request fills that slot (Friday); the Action utterance confirms the goal has actually been executed by the system; and the final Acknowledgement signals that the user is satisfied and the task can be closed. By tracking acts in this way, the agent knows exactly when information is missing (after a Request with an empty slot), when to ask a clarifying Question, and when the dialogue can be safely terminated (after an Action is followed by Acknowledgement with no further Requests).

Q4. Natural Language Generation and Surface Realization — NLG Pipeline

Problem Understanding

Input: a structured semantic representation with roles Action=Submit, Agent=Student, Object=Assignment, Recipient=Professor, Tense=Future, Time=Tomorrow. Output: a grammatically correct natural-language sentence expressing this meaning.

Algorithm (Pseudocode) — Pipeline Stages

ALGORITHM GenerateSentence(SemanticRep R)

1. Roles <- IdentifySemanticRoles(R) // Step 1: role identification
// {Action, Agent, Object, Recipient, Tense, Time}
2. Lexemes <- LexicalSelection(Roles) // Step 2: choose words
// Action->"submit", Agent->"student", Object->"assignment",
// Recipient->"professor", Time->"tomorrow"
3. SyntaxTree <- DetermineSyntacticStructure(Roles)
// Step 3: Subject-Verb-Object-PP structure
// S -> NP(Agent) VP(Verb NP(Object) PP(to Recipient) ADV(Time))
4. SyntaxTree <- ApplyTenseAndAgreement(SyntaxTree, Tense="Future")
// Step 4: verb "submit" -> "will submit"; agreement checked
5. Sentence <- SurfaceRealize(SyntaxTree) // Step 5: linearize + add
// determiners/prepositions/capitalization/punctuation
6. RETURN Sentence

Generated Sentence

"The student will submit the assignment to the professor tomorrow."

Explanation of the Transformation

- Semantic roles → syntax: Agent becomes the subject NP ("the student"), Action becomes the main verb, Object becomes the direct-object NP ("the assignment"), Recipient becomes a prepositional phrase ("to the professor"), and Time becomes a sentence-final adverbial ("tomorrow").
- Lexical selection maps each abstract role value to a concrete English word (Submit → "submit", Student → "student").
- Tense and agreement rules convert the base verb "submit" into the future form "will submit" and ensure subject-verb agreement (singular "student" with "will submit", which does not inflect for number, so agreement holds trivially in the future tense).
- Surface realization adds function words ("the", "to"), fixes word order, and applies capitalization and final punctuation to produce a fluent sentence.

Validation

- Grammaticality: subject-verb-object-prepositional phrase order is correct and the future-tense auxiliary "will" is properly placed before the base verb.
- Semantic consistency: every semantic role in the input (Agent, Action, Object, Recipient, Time) appears exactly once and correctly in the output sentence, with no added or missing meaning.
- Naturalness: the sentence reads as a fluent, native-sounding English sentence rather than a literal or awkward role-by-role concatenation.

Q5. Hybrid Machine Translation — Translation Workflow Algorithm

Problem Understanding

Input: the source sentence "The bank approved the student's loan application." Output: an accurate target-language translation, using a hybrid pipeline that combines an Interlingua (language-independent semantic) representation, statistical scoring of candidate translations, and contextual disambiguation of ambiguous words (e.g., "bank" = financial institution vs.

river bank).

Algorithm (Pseudocode)

```
ALGORITHM HybridTranslate(SourceSentence S)
1. ParseTree <- SyntacticParse(S)           // Step 1
2. SemanticStructure <- SemanticRoleLabel(ParseTree)
   // Agent=bank, Action=approve, Object=loan application,
   // Possessor(Object)=student
3. Interlingua <- BuildInterlinguaRepresentation(SemanticStructure)
   // Step 2: language-neutral frame:
   // APPROVE(agent=BANK_FINANCIAL,
   // object=LOAN_APPLICATION(owner=STUDENT))
4. AmbiguousWords <- DetectAmbiguousLexemes(S) // Step 3
   // "bank" -> {financial institution, river bank}
5. FOR each ambiguous word W in AmbiguousWords:
6.   Candidates <- GenerateCandidateTranslations(W, Interlingua)
7.   FOR each candidate C in Candidates:
8.     score(C) <- alpha*StatisticalCooccurrenceScore(C, S)
                  + beta*ContextualFitScore(C, Interlingua)
                  + gamma*DomainPlausibility(C, "finance")
9.   BestSense(W) <- ArgMax(Candidates, score) // Step 4
10. TargetSentence <- GenerateFromInterlingua(Interlingua, BestSenses)
    // Step 5: realize using target-language grammar
11. FinalTranslation <- SurfaceRealize(TargetSentence) // Step 6
12. RETURN FinalTranslation
```

Applying the Algorithm to the Example

- Step 1 — Parse: identifies "the bank" as subject/agent, "approved" as the main verb, and "the student's loan application" as the object with a possessive relation to "student".
- Step 2 — Interlingua: builds an abstract frame APPROVE(agent = BANK, object = LOAN_APPLICATION (owner = STUDENT)) that is independent of English or the target language's surface grammar.
- Step 3 — Ambiguity detection: "bank" is flagged as ambiguous (financial institution vs. river bank) and multiple candidate target-language words are generated for each sense.

- Step 4 — Statistical/contextual scoring: because the surrounding context contains "loan application" and "student," the contextual-fit and domain-plausibility scores strongly favour the financial-institution sense of "bank" over the river-bank sense.
- Step 5–6 — Final translation is generated from the disambiguated Interlingua frame and realized with correct target-language word order, e.g. (illustratively, for a generic target language): "[Financial-institution-word-for-bank] approved [student]'s [loan application]."

How the Components Work Together

The Interlingua representation supplies a language-independent semantic skeleton that keeps track of who did what to whom, which prevents the system from losing meaning across languages. Contextual information (the presence of "loan application" and "student") narrows down which sense of an ambiguous word like "bank" is intended, while statistical scoring quantitatively ranks the candidate translations using corpus co-occurrence evidence. Together, the semantic frame ensures structural correctness, the contextual score resolves lexical ambiguity correctly, and the statistical score selects the most fluent and natural-sounding target phrasing — improving both the accuracy (correct sense of "bank") and fluency (natural target sentence) of the final translation.

